SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1708**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

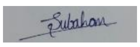
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I N.Kameshwar reddy(192525272) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right
ii. Cost of each move = 1 iii. Cannot pass through obstacles iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1) Backtracking Search – Doctor Shift Assignment (10 Marks)

Given

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints

D1 cannot work Night.

D2 must work before D3 (M < A < N).

Only one doctor per shift.

D3 cannot work Morning.

Every doctor gets exactly one shift.

Step 1: Assign D1

Possible shifts = Morning, Afternoon (Night not allowed)

Choose D1 = Morning

Current Assignment:

D1 → Morning

Available shifts:

Afternoon

Night

Step 2: Assign D2

Try Night

Current Assignment:

D1 → Morning

D2 → Night

Remaining:

D3 → Afternoon

Constraint Check:

D2 must be before D3.

Night is after Afternoon □

Backtrack

Try Afternoon

Current Assignment:

D1 → Morning

D2 → Afternoon

Remaining:

D3 → Night

Constraint Check:

D3 not Morning □

D2 before D3 (Afternoon before Night) □

One doctor per shift □

All assigned □

Solution Found

Backtracking Tree

```
             Start
               |
        D1 = Morning
          /       \
```

D2 = Night   D2 = Afternoon
□ |
D3 = Night
□
Final Valid Schedule
Shift
Doctor
Morning
D1
Afternoon
D2
Night
D3
Answer:
Morning → D1
Afternoon → D2
Night → D3

2. BFS Search Algorithm (Robot Grid) (10 Marks)
Assume the following 5 × 5 Grid
S . . X .
. X . X .
. X . . .
. . X X .
. . . . G
Legend:
S = Start
G = Goal
X = Obstacle
. = Free Cell
Coordinates:
Start = (0,0)
Goal = (4,4)
Movement:
Up
Down
Left
Right
Cost = 1
Manhattan Distance Formula
Goal = (4,4)
BFS Node Expansion
Step 1
Expand:
(0,0)

Queue:
[(1,0),(0,1)]
Step 2
Expand
(1,0)
Queue
[(0,1),(2,0)]
Step 3
Expand
(0,1)
Queue
[(2,0),(0,2)]
Step 4
Expand
(2,0)
Queue
[(0,2),(3,0)]
Step 5
Expand
(0,2)
Queue
[(3,0),(1,2)]
Step 6
Expand
(3,0)
Queue
[(1,2),(4,0),(3,1)]
Step 7
Expand
(1,2)
Queue
[(4,0),(3,1),(2,2)]
Step 8
Expand
(2,2)
Queue
[(4,0),(3,1),(2,3)]
Step 9
Expand
(2,3)
Queue
[(4,0),(3,1),(2,4)]
Step 10
Expand
(2,4)
Queue

[(4,0),(3,1),(3,4),(1,4)]
Step 11
Expand
(3,4)
Queue
[(4,0),(3,1),(1,4),(4,4)]
Goal Reached.
Optimal Path
S
↓
(1,0)
↓
(2,0)
↓
(3,0)
→
(3,1)
↓
(4,1)
→
(4,2)
→
(4,3)
→
G
Total Cost
Number of moves = 8
Final Answer
Algorithm Used: Breadth-First Search (BFS)
Optimal Path:
S → (1,0) → (2,0) → (3,0) → (3,1) → (4,1) → (4,2) → (4,3) → G
Total Cost = 8

3(a) Problem Understanding & Constraint Analysis

The autonomous rescue robot operates in a **partially observable** disaster environment where it must safely navigate from the **Start (S)** to the **Goal (G)** while minimizing the total movement cost. The constraints and their effects are:

1. **Movement Constraint (Up, Down, Left, Right)**
   - The robot can move only in four directions.
   - Diagonal movement is not allowed, reducing the number of possible paths.
2. **Obstacle Constraint**
   - Some cells are blocked and cannot be crossed.
   - The robot must detect these obstacles and find an alternative safe path.
3. **Limited Sensing Capability**
   - The robot can observe only adjacent cells.
   - It cannot know the complete environment initially and must explore while moving.

4. **Movement Cost**
    - Every normal move costs **1**.
    - The robot should avoid unnecessary movements to reduce the total cost.
5. **Risky Zones**
    - Entering a risky zone adds an additional cost of **2**.
    - The robot should avoid risky areas whenever a safer path exists.
6. **Dynamic Knowledge Update**
    - As the robot moves, it updates its map using newly observed information.
    - It replans the path whenever new obstacles or risks are discovered.

Decision-Making Process

- Sense nearby cells.
- Identify safe and blocked paths.
- Calculate movement costs.
- Choose the least-cost safe path.
- Update the map after every move.
- Repeat until reaching the survivor.

3(b) Solution Development (25 Marks)

Search Strategy: Uniform Cost Search (UCS)

Steps:

1. Start from node **S**.
2. Insert **S** into the priority queue with cost **0**.
3. Remove the node with the lowest cumulative cost.
4. Explore all valid neighboring cells.
5. Ignore blocked cells.
6. Add movement cost (1) and risky zone cost (+2 if applicable).
7. Update the priority queue.
8. Repeat until Goal **G** is reached.
9. Return the least-cost path.

Why UCS?

- Finds the optimal path.
- Considers different movement costs.
- Suitable for disaster environments with risky areas.

3(c) Application of AI Concepts

1. Intelligent Agent

- Uses sensors to observe nearby cells.
- Processes information and selects actions.
- Uses actuators to move.
- Continuously updates knowledge while navigating.

2. Rationality

- Chooses actions that maximize safety.
- Minimizes total travel cost.
- Avoids blocked and risky areas whenever possible.

3. Environment Type

- **Partially Observable:** Robot cannot see the whole environment.
- **Dynamic:** Environment may change during operation.
- **Sequential:** Current decisions affect future actions.
- **Discrete:** Grid consists of separate cells.
- **Single-Agent:** Only one rescue robot performs the task.

3(d) Justification

Uniform Cost Search is the most suitable algorithm because:

- It guarantees the **minimum-cost path**.
- It handles **different movement costs** effectively.
- It avoids unnecessary risky zones.
- It works well in partially observable environments by allowing replanning.
- It ensures safe and efficient rescue operations.

Conclusion:

The rescue robot behaves as an intelligent and rational agent by sensing its surroundings, updating its knowledge, avoiding obstacles and risky areas, and using **Uniform Cost Search** to reach the survivor with the **minimum total cost**, making it the best solution under the given constraints.